B3.2 Transport in plants (HL)

What the Cell · DP Biology · note-taking sheet · HL statements B3.2.17–18

Name _____ **Date** _____

How to use this sheet: go through the slides and videos with it, then close them and write from memory. Use your own words and add page numbers from the book (pp. 321–325) where you want more detail.

Guiding questions. Which adaptations help animals and plants move fluids around? How does transport in animals compare with transport in plants?

B3.2.17 Root pressure

Write the steps from mineral ions being pumped into the xylem to sap being pushed upwards. Which step needs ATP?

When does root pressure matter more than the pull from transpiration? Give two situations. What is guttation?

B3.2.18 Phloem

Draw a sieve tube element beside its companion cell. Label the sieve plate, plasmodesmata, the companion cell nucleus and mitochondria.

Give the job of each adaptation: sieve plates, little cytoplasm, no nucleus, many mitochondria in companion cells, plasmodesmata, thick rigid walls.

Define translocation, source and sink, with one example of each. Why does sap flow from source to sink?

Before you finish

Tick the ones you can explain without notes. Circle anything to ask about.

- ☐ Root pressure
- ☐ Guttation
- ☐ Sieve tube elements
- ☐ Companion cells
- ☐ Sources and sinks
- ☐ Translocation